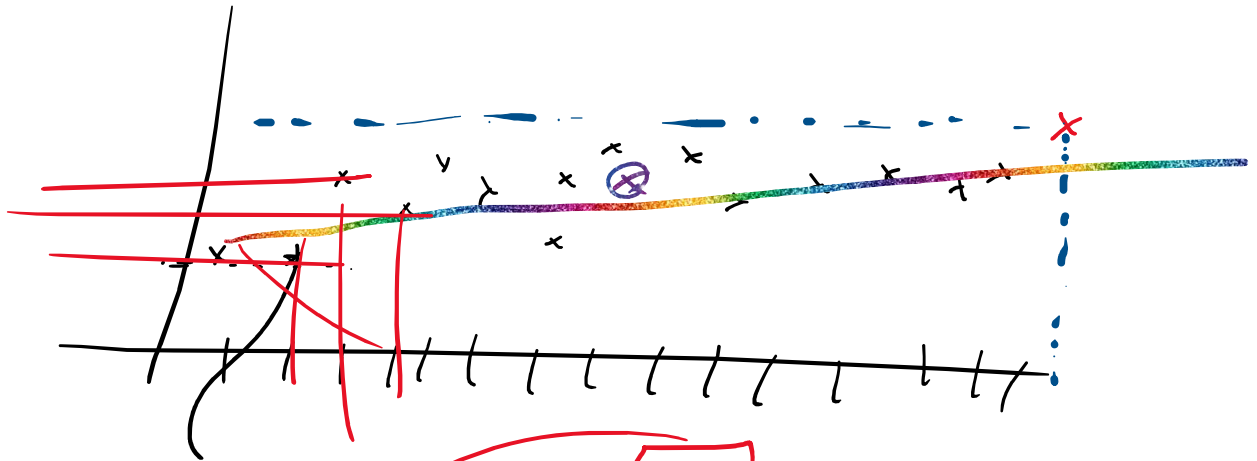


Module 06: Linear Regression:



$$y = mx + b$$

$$y = a + bx$$

Parameters

Dependent Variable

Regression Co-efficient

Slope

Independent Variable

$\Rightarrow$

$$a = \bar{y} - b \cdot \bar{x}$$

$$b = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{SS_{xy}}{SS_x}$$

$\Rightarrow$ (x) Heights (cm) (y) Weights (kg)

158	60	172	68	182	75	189	80	190	85
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g) weights (x_j) | 60 68 79 80 85

Supervised Learning: Questions & Answers

$$y = a + bx$$

$$a = \bar{y} - b \cdot \bar{x}$$

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

x	y	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$
158	60	158 - 178.2 -20.2	408.04	-14.4	290.88
172	68	172 - 178.2 -6.2	38.44	-6.4	39.68
182	79	182 - 178.2 3.8	14.44	4.6	17.48
189	80	189 - 178.2 10.8	116.64	5.6	60.48
190	85	190 - 178.2 11.8	139.24	10.6	125.08
			716.8		533.6

$$\bar{x} = 178.2$$

$$\bar{y} = \underline{\underline{74.4}}$$

$$b = \underline{\underline{\sum (x_i - \bar{x})(y_i - \bar{y})}}$$

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$b = \frac{533.6}{716.8} = \boxed{0.744}$$

$$a = \bar{y} - b \cdot \bar{x}$$

$$= 74.4 - (0.744 \times 178.2)$$

$$a = \boxed{-58.18}$$

$\therefore$ Linear Regression Equation is as follows:

$$\hat{y} = -58.18 + (0.744) \cdot x$$

$$\therefore \hat{y} = \underline{\underline{a + bx}}$$

Assume the weight of the person with height
as 210 cm.

$$\hat{y} = -58.18 + (0.744) \times 210$$

$$\hat{y} = 98.07$$

$\therefore$ Weight of the person predicted

is 98 kgs & 07 grams.

$\Rightarrow$ Accuracy of Model : Goodness of Fit :

R^2 - Score

$$\text{Formula } R^2 = \frac{SSR}{SST} = 1 - \left(\frac{SSE}{SST} \right)$$

$\Rightarrow$ SSE : Sum of Squared Errors

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$\Rightarrow$ SSR = Sum of Squared Regressor.

$$\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$\Rightarrow SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$\Rightarrow \hat{y} = -58.18 + (0.744)x$$

X	y	$\hat{y}$	$\bar{y}$	$(\hat{y}_i - \bar{y})$	$(\hat{y}_i - \bar{y})^2$
→ 158	60	59.37	74.4	-15.03	225.90
172	68	69.78	74.4	-4.62	21.34
182	75	77.2	74.4	2.82	7.95
189	80	82.43	74.4	8.03	64.48
170	85	83.18	74.4	8.7	75.69
					<u>395.25</u>
					SSE

$$SST = \underline{413.2}$$

$$R^2 = \frac{SSR}{SST} = \frac{395.25}{413.2} = \underline{0.95}$$

95% Accuracy of model